pfSense Installation on a Virtual Machine

introduction : The objective of this lab is to deploy pfSense as a virtual firewall and implement a Zero-Trust approach using a Default-Deny policy to secure the internal network.

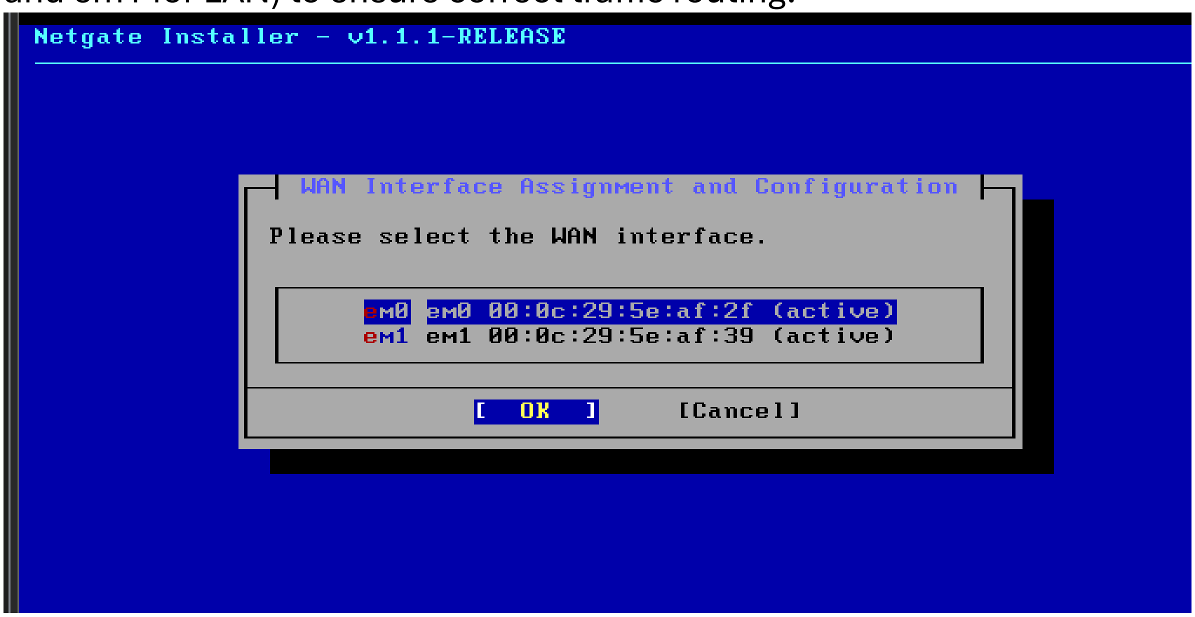
Installation Steps : In this section, the primary steps for deploying pfSense on a virtualized environment are documented:

- **Step 1:** Hypervisor Configuration

Creating a new Virtual Machine on VMware Workstation with the appropriate hardware specifications (CPU, RAM, and Storage) and adding two Network Adapters to represent WAN and LAN segments.

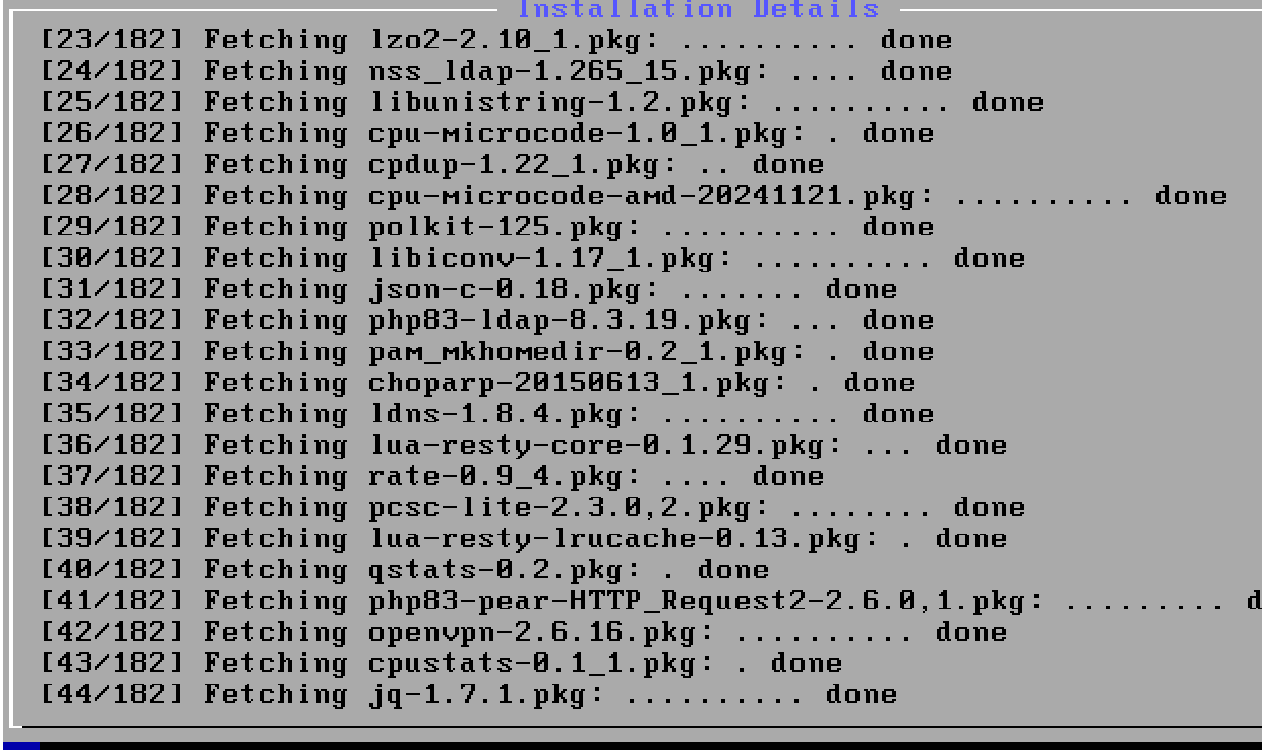
- **Step 2:** Interface Assignment (Boot Phase)

Upon booting the pfSense ISO, the system detects the virtual network cards. As shown in the screenshots, the interfaces were assigned (e.g., em0 for WAN and em1 for LAN) to ensure correct traffic routing.



- **Step 3:** Package Fetching and System Installation

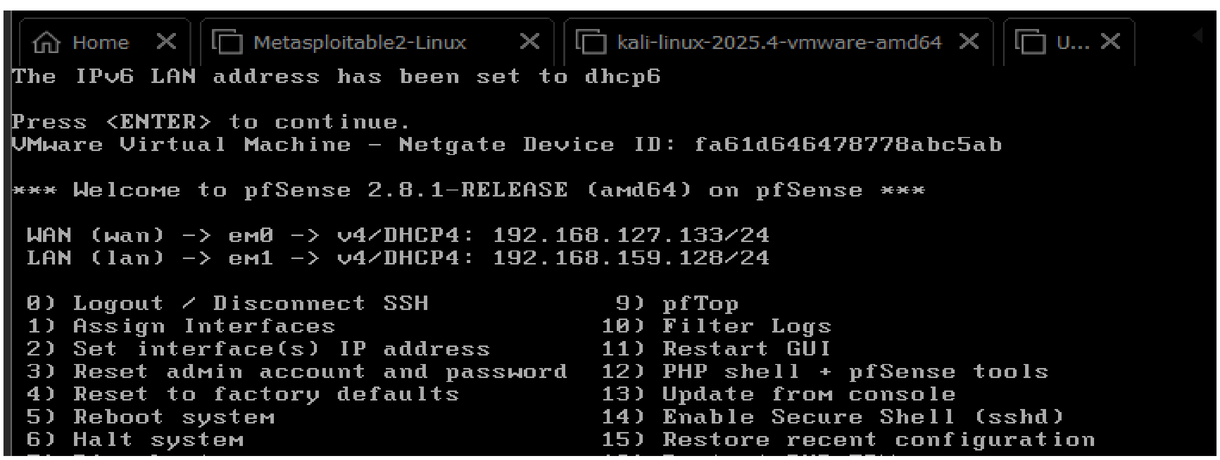
The installer proceeds to format the virtual disk and fetch the necessary core packages. This step confirms the integrity of the installation media and the successful extraction of the FreeBSD-based kernel.



- **Step 4:** Initial Console Configuration

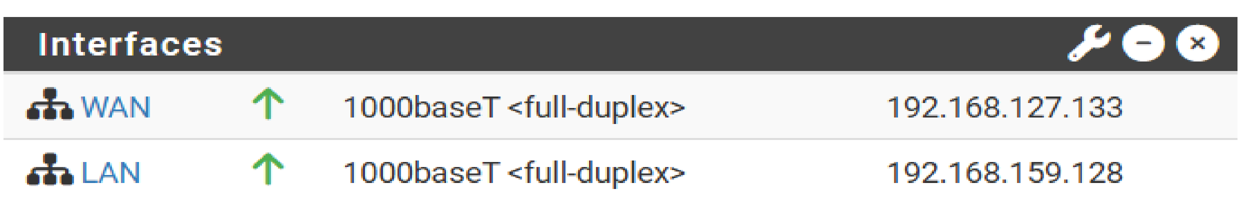
After the reboot, the pfSense console menu appeared, displaying the assigned IP addresses for both interfaces.

- WAN IP: Obtained via DHCP from the host network.
- LAN IP: Configured to serve the internal protected network (192.168.159.128/24).



- **Step 5:** Accessing the WebGUI

The final installation proof was achieved by accessing the pfSense Dashboard through a web browser from a host within the LAN segment, confirming that the web configurator is fully operational.



Network Interface Configuration :

This section summarizes the network interface assignments and the logic behind the addressing scheme used in this lab. For visual verification of these settings, please refer to the screenshots in the previous section.

• WAN Interface (em0):

The external interface was configured to dynamically obtain an IP address via DHCP (192.168.127.133). This setup allows the pfSense firewall to communicate with the outside world and provides the necessary internet gateway for internal clients.

• LAN Interface (em1):

The internal protected interface was assigned a static IP address of 192.168.159.128/24. This address serves as the default gateway for all devices within the local network, ensuring a controlled and isolated environment.

• Connectivity Rationale:

By separating the network into WAN and LAN segments, we establish a clear security boundary. This physical and logical separation is essential for implementing the firewall rules that will govern traffic flow in the following sections.

Firewall Ruleset Design and Justification :

In this section, the security policy was implemented based on the Principle of Least Privilege. The strategy focuses on blocking all non-essential traffic and only permitting specific services required for network operations.

• Default-Deny Strategy:

A "Default-Deny" policy was established on both WAN and LAN interfaces. This ensures that any traffic not explicitly defined in the allowlist is automatically dropped, significantly reducing the attack surface.

• Inbound Policy (WAN):

All inbound traffic from the external network is blocked to protect the internal infrastructure from unauthorized access or external probes.

• Outbound Policy (LAN):

Instead of allowing full internet access, only specific ports were opened:

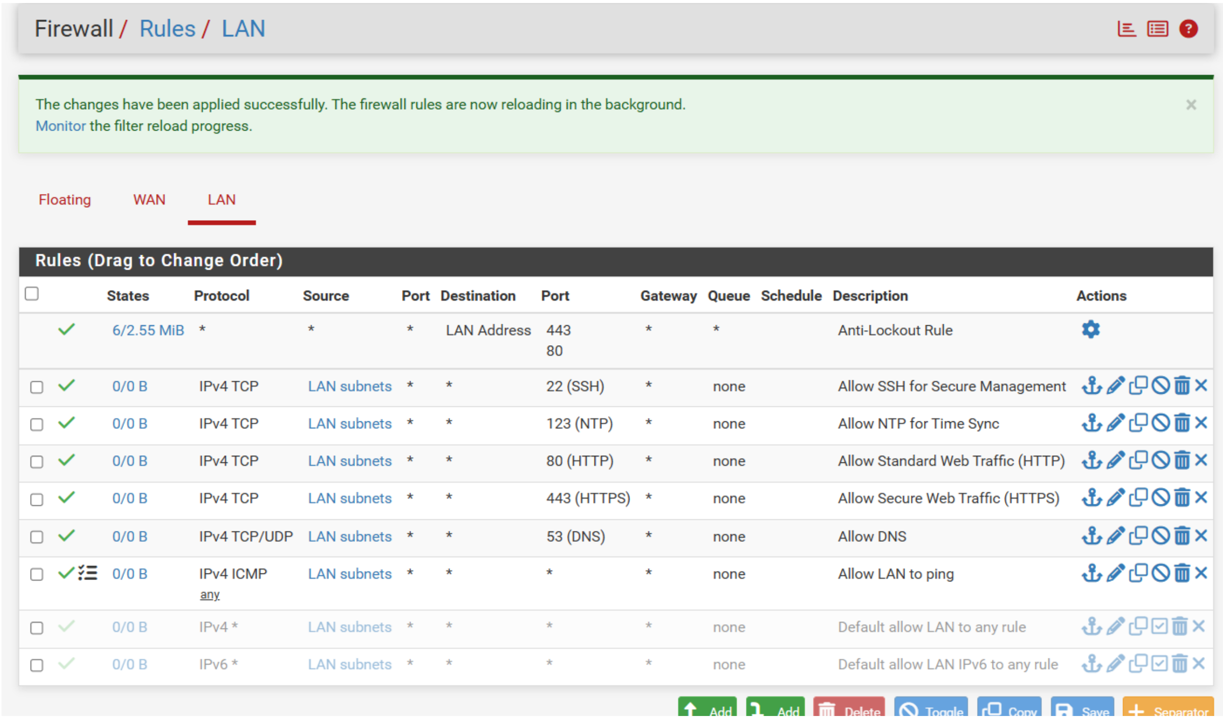
• DNS (53): For domain name resolution.

• HTTP/HTTPS (80/443): For secure web browsing.

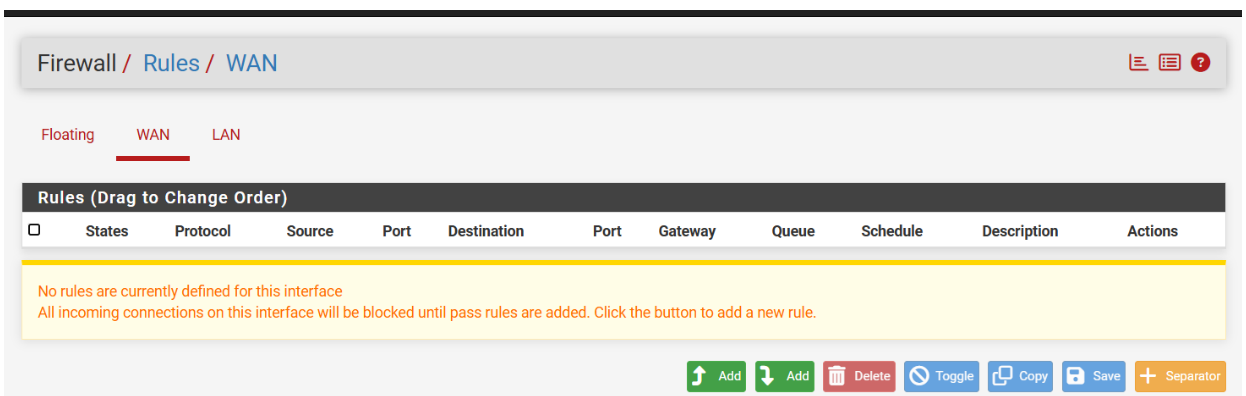
• SSH (22): For secure remote administration.

• NTP (123): For time synchronization.

• ICMP: For internal network diagnostics.



LAN Interface Ruleset implementing the Allowlist.



WAN Interface Ruleset enforcing Default-Deny for inbound traffic.

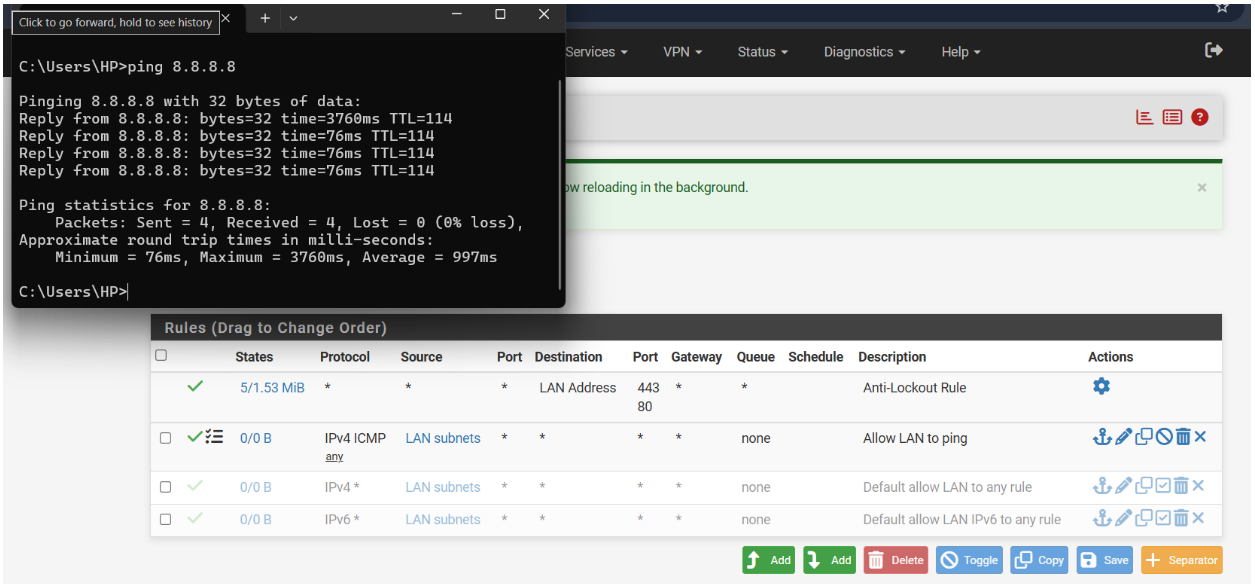
Rule Table Summary:

The following table provides a detailed summary of the firewall rules implemented on the pfSense interfaces. It outlines the protocols, port assignments, and the technical justification for each rule to ensure a secure and functional network environment.

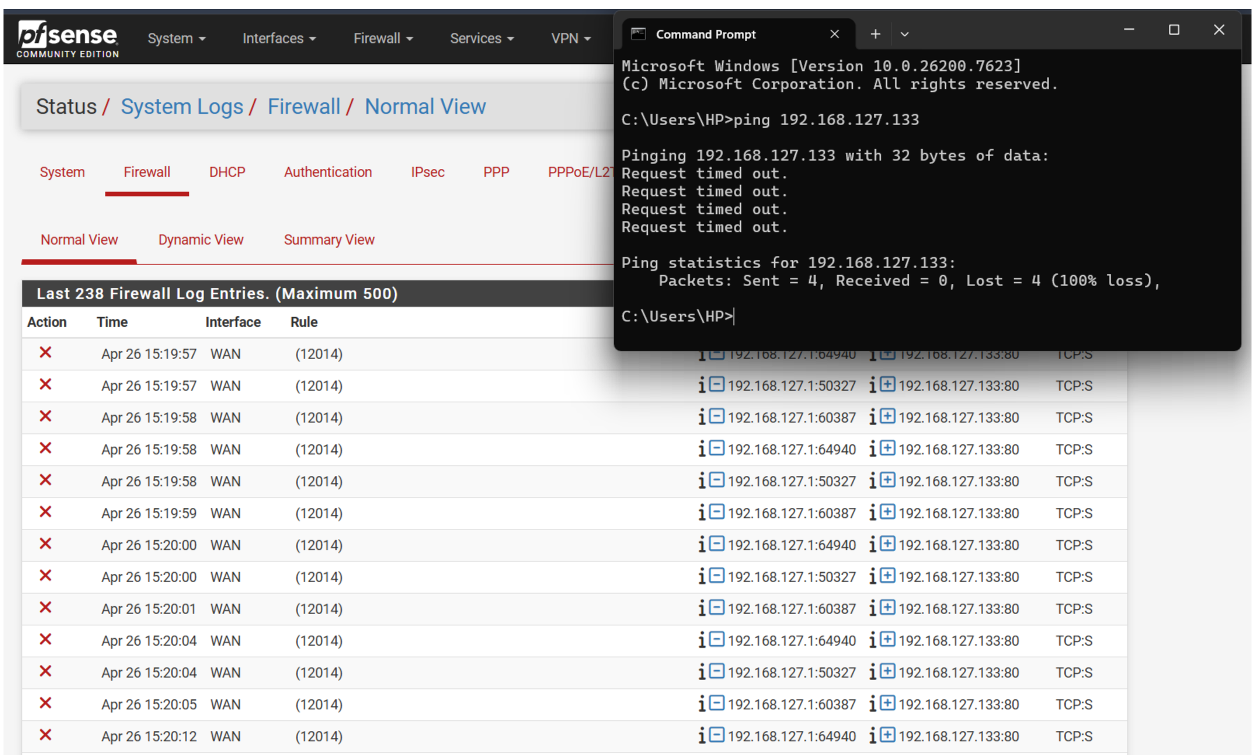
Interface	Protocol	Source	Destination	Port	Action	Justification
LAN	*	*	LAN Address	"443,80"	pass	Anti-Lockout Rule: Ensures administrative access to the web configurator.
LAN	IPv4 TCP	LAN subnets	*	22 (SSH)	Pass	SSH Management: Allows secure remote CLI access restricted to internal network.
LAN	IPv4 TCP	LAN subnets	*	123 (NTP)	Pass	Time Sync: Enables local clients to synchronize system clocks.
LAN	IPv4 TCP	LAN subnets	*	80 (HTTP)	Pass	Web Traffic: Permits standard unencrypted web browsing for internal hosts.
LAN	IPv4 TCP	LAN subnets	*	443 (HTTPS)	Pass	Secure Web: Allows encrypted web traffic (SSL/TLS) for secure browsing.
LAN	IPv4 TCP/UDP	LAN subnets	*	53 (DNS)	Pass	DNS Resolution: Allows clients to resolve domain names.
LAN	IPv4 ICMP	LAN subnets	*	*	Pass	Network Diagnostics: Permits the use of Ping for troubleshooting.
WAN	Any	Any	Any	Any	Block	Default Deny: Blocks all unsolicited inbound traffic to ensure security.

Testing and Verification:

To validate the firewall configuration, several tests were conducted to ensure that the ruleset correctly permits authorized traffic while blocking unauthorized access.



The successful ping confirms that ICMP and DNS rules are working.



Logs show the Default-Deny policy blocking unauthorized WAN traffic.

Conclusion:

This lab demonstrated the successful deployment of a pfSense firewall. By implementing a "Least Privilege" approach, the network's security posture was significantly enhanced. The testing phase confirmed that all critical services are operational while maintaining a robust defense against unauthorized access.